

# A Mutualistic Fungus of Tall Fescue Grass

Mutualism is an interaction between two species in which both benefit. A classic example of this relationship is between tall fescue grass and a fungus named *Epichloë coenophiala*. That fungus is completely internal and grows intercellularly (between the cells) in the above-ground portion of the grass. The grass supplies all the nutritional needs of the fungus. Infected plants exhibit no external symptoms of the fungus and no disease symptoms. In contrast, pathogenic (disease-causing) fungi result in symptoms that include wilt, leaf spots, deformed plant structures, or death of plant tissue. Pathogenic fungi often invade plant host cells and absorb the cells' contents (thereby eventually killing the cells) or produce chemicals that kill them outright.

Tall fescue is extensively used as both a pasture grass and a turf or lawn grass. It is a hardy, vigorously growing grass but has been associated with various health problems in cattle and horses. Beef cattle often show reduced weight gains when grazing on tall fescue pastures, and dairy cattle experience lower milk yields. Female horses may be more likely to lose their offspring during pregnancy and may not produce milk following a live birth. Reports of such adverse effects resulted in microscopic investigation of leaf and stem tissue that revealed the presence of the *Epichloë coenophiala* fungus. Surveys of infected pastures found a positive correlation between the percentage of infected plants and the severity of animal symptoms. The fungus in the grass was found to produce various alkaloids (toxic nitrogen-containing organic molecules). One type of alkaloid (ergot alkaloids) contributes to the livestock's symptoms and reduced grazing, whereas another type (looline alkaloids) makes the grass more resistant to insects. Interestingly, the insect-deterring alkaloids are not produced when the fungus is grown in the laboratory apart from the plant. Because these chemicals defend the grass as well as the fungus from consumption by both livestock and insects, the mutualistic relationship has been termed defensive mutualism.

Because the *Epichloë coenophiala* fungus is not external on the plant, the usual mechanism of spread by fungi does not take place. Typical fungi form spores (reproductive units consisting of one or more cells that are similar to the seeds of plants). Fungal spores spread through the air to susceptible plants and usually

require an extended period of moist conditions for their survival and germination (the beginning of growth of a spore or seed), and for the subsequent infection of the host plant. The tall fescue fungus, in contrast, penetrates the flower head and infects the plant embryos in the developing seeds. Therefore, seeds collected from infected plants result in infected seedlings following germination. As the plant grows, the fungus grows along with it by division and stretching of the hyphae (threadlike structures that make up the body of a fungus) at the base of the grass, where plant cell division also occurs. Thus, growth of the fungus is concurrent with growth of the plant, which ensures that new stems and leaves of the grass are infected with the fungus. Fungus-free tall fescue plants can result if infected seed is stored for several months before planting. The fungus usually dies out during most long-term storage conditions.

The alkaloids produced by the fungus also affect microorganisms in the immediate environment of the plant. Nondisease-causing bacteria, recovered from the surfaces of fungus-infected tall fescue plants, can use loline alkaloids, released on the leaves, as a source of nutrients. The population size of these bacteria is about eightfold greater on infected plants than on fungus-free tall fescue plants or on plants colonized by related fungi incapable of producing loline. Thus, many components of the fungus-infected tall fescue community are impacted by the fungus: the grass itself, animals, insects, and microbes.

Although the fungus of tall fescue is detrimental to livestock, it is considered beneficial when grass is grown for other purposes. Insect pests are numerous in turf (short, thick grass and the soil it grows in), and tall fescue's increased resistance to them is advantageous when grass is grown for house lawns or sports fields. It may be possible to develop pasture varieties of tall fescue that are infected with this fungus for the purpose of insect resistance, but with the deletion of the fungal genes for the toxins that harm livestock.